

Ethan Hopkins

Software Engineer

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Glendale, New York

Analytical Software Engineer specializing in high-availability architecture and legacy modernization. Combines a B.S. in Computer Science with 10 years of high-stakes operational leadership to build scalable, mission-critical systems that drive business impact.

Work Experience

Software Engineer

Dec 2025 - Jun 2026

wyoming.com | Riverton, WY

Engineering mission-critical, zero-downtime infrastructure and modernizing legacy ISP systems through high-availability database clustering and AI-accelerated development with an emphasis on accuracy and safely working in production systems.

- Improved company's CRM transition by transforming insecure Perl scripts into a secure Python/Django API framework, while ensuring zero-downtime for crucial 911/VoIP services during database migrations. Utilized Prometheus for realtime metrics tracking.
- Orchestrated and performed database administration for a MariaDB high-availability cluster using primary-to-primary ring replication to eliminate single points of failure; architected an enterprise Grafana/Loki/Ansible observability stack integrated with LDAP for secure, role-based monitoring.
- Unified and streamlined internal tools into a Django/Debian ecosystem using Celery and Redis, enhancing maintainability, security documentation, and deployment reliability across multiple teams. Enhanced front-end design to further user experience.
- Authors enterprise version control policies, supervises AI-assisted coding workflows, and mentors the engineering team on SQL optimization while translating complex requirements from Sales and Accounting into actionable system designs.
- Developed and executed machine learning experiments and prototyping to validate hypotheses and drive innovation in software development processes.
- Integrated Automated Testing into the CI/CD pipeline to ensure continuous delivery of high-quality software via GitHub Actions.

Retail Manager

Oct 2017 - Nov 2025

Walmart | Wyoming (Multiple Cities)

Directed \$130M+ annual customer-facing operations and 350+ associates while engineering regional automation tools and secure digital asset ledgers to drive \$300k+ in business impact. Developed high-impact communication skills

- Managed a \$130M revenue facility and 350+ staff, yielding a \$300k+ yearly impact via strategic revenue operations, cross-functional collaboration, and crisis leadership, ensuring operational stability and morale.
- Assisting marketing efforts by executing social media driven outreach and store initiatives.
- Engineered and deployed regional VBA automation tools and a custom MS Access security database while spearheading technical training for 300+ staff to bridge the gap between corporate technical requirements and ground-level execution. Uses Salesforce for data collection.

Projects

Predicting Uber Ride Cancellations

Present

Engineered a predictive classification model using Python and scikit-learn to identify ride cancellation risks, implementing hyperparameter tuning and data preprocessing to optimize precision for real-time revenue reclamation.

Core Skills

Languages: Python, Java, SQL, Ansible, HTML, CSS, JavaScript, VBA, GML

Databases: MySQL, MariaDB, PostgreSQL, Microsoft Access

Frameworks: Django, Spring Boot, Angular, Node.js, Celery, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib

AI Coding: GitHub Copilot, Gemini, Claude Code

Tools & DevOps: Git, GitHub, GitLab, Docker, VS Code, IntelliJ IDEA, Unity, GameMaker Studio, Google Apps, Prometheus, Agile

Soft Skills: Resourceful, Effective communication

Education

Western Governors University Sep 2025

Bachelor of Science Computer Science

Minor: Data Structures and Algorithms, Advanced Database Management, Artificial Intelligence, Mathematics

Purdue University Global May 2021

Bachelor of Science Business Administration

Certificates

Linux Essentials

LPI

ITIL Foundation 4